

12

FEATURE

FEATURE 12 OF 17 - TEST IT YOURSELF, ANYWHERE

Click Map

Click any one of 24,000 cells in the basin and get the model's full profile underneath it.

PORT	PAGE FILE	CELLS	DEPTHS
8512	app/phase2/clickmap_page.py	24,000, all clickable	15 per cell

This is the whole claim of the project reduced to one interaction. An Argo float answers 'what is it like at 500 metres' only where a float happens to be, every five to ten days. This answers it everywhere, every day - and it is the only page where a viewer can test that themselves.

1. What this feature is, in plain words

A map of the basin at a chosen depth. Every cell is clickable. Click one and the model's full 15-depth profile for that exact cell appears beside it, with its uncertainty band.

The profile is taken from the map's own array rather than re-running the model. That is deliberate: a second prediction call would cost another inference and, worse, would let the number in the side panel drift away from the colour under the cursor if the two paths ever disagreed. A test pins the field and point paths to be identical, precisely so this shortcut is sound.

There are three kinds of blank on this map and they are not the same thing. A cell can be land; it can be ocean whose sea floor sits above the depth you asked for; or it can be outside the grid. Each says something different, and the page names which one you clicked instead of going silent.

THE EVERYDAY PICTURE

Most ocean data is a scatter of pinpricks - wherever an instrument happened to be. This is the same information as a continuous surface you can probe anywhere, and it invites the audience to try to catch it out.

2. What you actually see on the screen

One map, one profile, and no place to hide:

- The full basin at a selected depth, drawn as 24,000 individually clickable cells.
- The clicked cell's complete profile - all 15 depths - with its calibrated plus-or-minus 2 sigma band.
- An explicit message when you click land: *that cell is land*, rather than a phantom profile.
- An explicit message when the sea floor is above your chosen depth.
- A GPU toggle, which is how a device bug that had existed since the parameter was written was finally discovered.

3. How to run it, step by step

RUN THIS

```
PYTHONPATH=src python -m streamlit run app/phase2/clickmap_page.py --server.port 8512
```

- 1 Run the command and open **http://localhost:8512**.
- 2 Choose a depth - 100 m is a good choice, because it is the hardest depth in the problem.

- 3 Invite a judge to click anywhere in the water. The profile appears immediately.
- 4 Now click on India, or on the Arabian peninsula. The page says **that cell is land**. Show this deliberately - it is the difference between a system that knows what it does not know and one that guesses.
- 5 Click a shallow shelf cell with the depth set to 1000 m to see the sea-floor message.
- 6 Compare the number in the side panel with the colour under the cursor, and explain that they come from the same array by construction.

FOUR BUGS THAT ONLY RENDERING COULD FIND

The cells first drew as a handful of enormous blocks; the basin drew at the wrong aspect ratio, so the Bay of Bengal was the wrong shape; the profile chart forced a zero baseline, squeezing a real 0.83 degC uncertainty band into 2% of the plot - an uncertainty band drawn so thin it reads as certainty; and land was silently dropped, so a click on India selected nothing while the panel still said 'click any cell'. **None of those was visible from a test. All four were found by opening the page.**

4. Why the problem statement needs this

The problem statement asks for a gridded, basin-wide reconstruction. This page is the direct demonstration that the output really is complete - not a handful of validated points, but every cell, every depth, ready to be interrogated at random.

It is also the page that proves the system knows its own boundaries. A model that produces a confident-looking profile for a point in the Thar Desert has learned nothing useful; this one refuses.

PS requirement	How this feature answers it
0.25 degree spatial resolution	All 24,000 cells of the 100 x 240 grid are individually addressable.
15 standard depths	Every clicked ocean cell returns all 15 levels, or names the reason it cannot.
Daily reconstruction	Any date in the 388-day bundle, and the map redraws for it.

5. The measured numbers behind it

Verified live in the browser on real data [VERIFIED]:

Test	Result
Ocean cell at 17.50 N, 57.50 E	15 of 15 levels returned - 28.27 degC at the surface falling to 9.11 degC at 1000 m , with a plus-or-minus 2 sigma band of 0.83 to 3.63 degC, calibrated.
Land cell at 21.75 N, 47.50 E	'That cell is land' - and no phantom profile.
Cell classification at 100 m	9,763 water + 2,069 below sea floor + 12,168 land = exactly 24,000 .
Cell classification at 1000 m	8,973 water + 2,859 below sea floor + 12,168 land - land unchanged, water strictly fewer, as physics demands.
Cross-check against an independent count	water + sea floor = 11,832 at both depths, matching the ocean-cell count recorded independently by the export timing measurement.

- The order of the two classification checks is load-bearing: asking 'is the value finite' before asking 'is it land' would label the entire coastline as being below the sea floor, which is a claim about bathymetry the data never made.

6. The impact - and who it is for

Who	What changes for them
The jury	An open invitation to try to break the system, which is the strongest possible statement of confidence.
Operational users	A point-query interface anyone can use without knowing anything about the model.
Fishers and mariners	The most direct route from 'where am I' to 'what is the water under me doing'.
The team	The page that found four rendering bugs no test could have caught, including one that made uncertainty look like certainty.

WHY SOCIETY GAINS FROM THIS

Subsurface ocean information has historically been available only to institutions with ships and floats. A map anybody can click, covering an entire basin every day, changes who is able to ask the question at all - which matters most for the small operators who were never going to have their own instruments.

7. Say this to the jury

Please click anywhere. That is the whole demonstration. An Argo float can tell you what 500 metres looks like only where a float happens to be, every five to ten days - this answers it at all 24,000 cells, every day. Now click on land. It says 'that cell is land', because at 100 metres our classification is 9,763 water plus 2,069 sea floor plus 12,168 land, which is exactly 24,000. Nothing is being quietly filled in.

Best used immediately after the accuracy numbers - it makes them tangible.

8. Questions the jury will ask

Q. Is the profile in the panel really the same as the colour on the map?

A. Yes, by construction - it is read from the same array rather than re-predicted, and a test pins the whole-field and single-point paths to be identical. A second prediction call would let the two drift apart, and we removed that possibility rather than monitoring it.

Q. What happens if I click somewhere impossible?

A. You get told which impossible thing it was. Land, sea floor above your chosen depth, or outside the grid - those are three different facts and the page names the right one. A bare blank would conflate them.

Q. How do you know your land mask is right?

A. It cross-checks three ways. The classification sums to exactly 24,000 at every depth; land stays constant as depth increases while water strictly decreases; and the ocean-cell total of 11,832 matches a count produced independently by a completely different part of the system.

9. The honest limit - say it before they find it

IT SHOWS THE MODEL, NOT THE TRUTH

Clicking a cell shows what the model says, not what the ocean is. The uncertainty band is the honest guide to how much to trust each value, and Feature 8 is where the same prediction is checked against a real float. Note also that the map draws one depth at a time - the vertical relationships between depths are better read in the transect (Feature 10) or the cube (Feature 4).

IN ONE LINE

Every one of 24,000 cells is clickable, and the three kinds of blank are named rather than left silent.

24,000

CELLS, AND THE CLASSIFICATION SUMS TO IT
EXACTLY

11,832

OCEAN CELLS, CROSS-CHECKED
INDEPENDENTLY

4 bugs

FOUND ONLY BY OPENING THE PAGE

Feature 12 of 17. Next: Feature 13.